

COMPSCI5079 (M)

Cryptography & Secure Development

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Lecture 2:

Traditional Encryption and Decryption schemes





Lecture Outline

1. Types of Attack

- Threats that need to be protected against.

2. Transposition and Substitution ciphers.

3. Homophonic Ciphers

- Plaintext letters can have several ciphertext letters.

4. Polyalphabet Ciphers and Rotor Machines.

- Mechanical encryption devices.

5. Running key Ciphers

- The key is the same length as the plaintext



Substitution Ciphers



Substitution Ciphers

- Substitution ciphers provide matching between letters in the plaintext alphabet to other letters in the same or a different alphabet for the ciphertext.
 - Two types: **Shift Substitution** and **Multiplicative Substitution**
- In the simple case, each English letter is replaced by another English letter.
- In more complex cases, English letters can be replaced by letters in another alphabet.
- There can even be more letters in the ciphertext alphabet than there are in the plaintext alphabet.
 - Some letters in the plaintext can be replaced by a choice of different letters in the ciphertext.



Shift Substitution

- This example assumes the letters are replaced by numbers:

a=0, b=1, c=2, d=3, e=4, f=5, g=6, h=7, i=8, j=9,
k=10, l=11, m=12, n=13, o=14, p=15, q=16, r=17, s=18,
t=19, u=20, v=21, w=22, x=23, y=24, z=25.

- A shift substitution shifts each letter of the alphabet along by a fixed amount, with a wrap-around at the end. The formula is:

- $c = (p + k) \% n$

where c is the cipher letter, p is the plain text, n is the size of the alphabet and k (i.e., the secret key) is the number of characters to shift.

- Decryption uses the inverse transformation

- $p = (c - k) \% n$



Example: Caesar Cipher

- An early example of a shift substitution is the Caesar cipher with $k = 3$
- This confused the Gauls during Caesar's wars.
 - As reported by Caesar in his own book!
- The Caesar cipher relied on a secret algorithm, which is usually a bad idea.
 - But it worked for Caesar.
- Diffusion is again poor. Why?
 - Changing one letter in the plaintext only changes one letter in the ciphertext.



Example: Caesar Cipher

securityandcryptography becomes
vhfxulwbdogfubswrjudskb

- The letter frequencies have been shifted.
- We can use the most frequent letters in the ciphertext to guess the shift.
 - For instance: if the most common letter in the ciphertext is **h**, we guess that it corresponds to a common letter such as **e**, with a shift of **3**.
 - This attack works better for longer messages.



Multiplicative Substitution

- A multiplicative substitution is slightly more sophisticated, and uses the formula:

$$c = (p \times k) \% n$$

- Decryption uses k' , the inverse of k

$$p = (c \times k') \% n, \text{ where } k k' = 1 \pmod{n}.$$

- A more complex version would combine shift and multiplication.

$$c = (p \times k1 + k2) \% n$$

- In this case, n should be a prime number, to make sure that all values of k can be used (they must have inverses).



Breaking the Cipher

- The Unicity Distance is quite small since there are not many possible keys.
 - There are 25 keys, assuming shifting with 0 or multiplying with 1 are not used.

$$H(K) = \log_2(25) \text{ and so } N_u = \log_2(25) / 3.2 = 1.5$$

- Single-letter frequencies can be used to break these codes quite easily.
- An exhaustive search (Brute Force) is also possible.
 - Shift and multiply have $25^2 = 625$ possible keys.



General English to English Substitutions

- The key is a substitution between letters of the alphabet.
 - There are a large number of different possible substitutions and so keys.
- The unicity distance calculation is similar to a transposition cipher, and
$$N_u = 27$$
- A “ciphertext-only attack” will then use single-letter frequencies to break the cipher.



Lecture Summary



Transposition and Substitution Ciphers

➤ Transposition:

- Split characters into blocks of fixed length d .
- Rearrange the characters inside a block according to a key-dependent permutation.
- Decryption uses the inverse permutation to recover the plaintext.

➤ Substitution

- Provide a matching between letters in the plaintext alphabet to other letters in the same or a different alphabet for the ciphertext.
- Shift Substitution
- Multiplicative Substitution